

Instructions: Use proper notation, and show all of your work to receive credit.

1. (2 pts) True or False (circle your answer): The Law of Iterated Expectations says that, for any two random variables X and Y , $E[E(Y|X)] = E(Y)$.

True

False

2. (6 pts) Let Y be a random variable with probability density function given by

$$f(y) = \begin{cases} 2(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

Find the density function of $U_3 = Y^2$.

$$F_{U_3}(u_3) = P(U_3 \leq u_3) = P(Y^2 \leq u_3) = P(-\sqrt{u_3} \leq Y \leq \sqrt{u_3}) \stackrel{\text{since } y \geq 0}{=} P(Y \leq \sqrt{u_3}).$$

Note: $0 \leq y \leq 1 \Rightarrow 0 \leq y^2 \leq 1$ (i.e., $0 \leq u_3 \leq 1$).

Case 1: $u_3 < 0$. $F_{U_3}(u_3) = 0$.

Case 2: $0 \leq u_3 \leq 1$.

$$\begin{aligned} F_{U_3}(u_3) &= P(U_3 \leq u_3) = P(Y^2 \leq u_3) = P(Y \leq \sqrt{u_3}) \\ &= \int_{-\infty}^{\sqrt{u_3}} f(y) dy \\ &= \int_{-\infty}^0 0 dy + \int_0^{\sqrt{u_3}} 2(1-y) dy \\ &= 2 \left[y - \frac{1}{2} y^2 \right]_0^{\sqrt{u_3}} \\ &= 2 \left(\sqrt{u_3} - \frac{1}{2} u_3 \right) \\ &= 2\sqrt{u_3} - u_3 \end{aligned}$$

Case 3: $u_3 > 1$. $F_{U_3}(u_3) = \int_0^1 2(1-y) dy = 1$.

$$\text{So, } F_{U_3}(u_3) = \begin{cases} 0, & u_3 < 0 \\ 2\sqrt{u_3} - u_3, & 0 \leq u_3 \leq 1 \\ 1, & u_3 > 1 \end{cases} \Rightarrow f_{U_3}(u_3) = \begin{cases} \frac{1}{\sqrt{u_3}} - 1, & 0 \leq u_3 \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

↑
differentiating

3. (4 pts) Suppose that Y has a gamma distribution with $\alpha = n/2$ for some positive integer n and β equal to some specified value. Use the method of moment-generating functions to show that $W = 2Y/\beta$ has a χ^2 distribution with n degrees of freedom.

Suppose $Y \sim \text{Gamma}(\alpha = \frac{n}{2}, \beta)$.

$$\begin{aligned} \text{Then, } m_W(t) &= E(e^{tW}) \\ &= E(e^{t(2Y/\beta)}) \\ &= m_Y\left(\frac{2t}{\beta}\right) \\ &= (1 - \beta(2t/\beta))^{-n/2} \\ &= (1 - 2t)^{-n/2} \\ &\quad \underbrace{\hspace{1cm}}_{\text{mgf of a } \chi_n^2 \text{ RV}} \end{aligned}$$

Hence, by uniqueness of MGFs, $W \sim \chi_n^2$.

4. (3 pts) Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed random variables such that for $0 < p < 1$, $P(Y_i = 1) = p$ and $P(Y_i = 0) = q = 1 - p$. Find the moment-generating function for Y_1 .

y_i	0	1
$p(y_i)$	$1-p$	p

$$\begin{aligned} m_{Y_1}(t) &= E(e^{tY_1}) = \sum_{y_i=0}^1 e^{ty_i} p(y_i) \\ &= 1 \cdot p(0) + e^t \cdot p(1) \\ &= 1-p + pe^t \end{aligned}$$

5. (4 pts) If $X \sim N(-2, 3)$ and $Y \sim N(3, 2)$ are independent, find the distribution of $W_1 = X - 2Y$. Be sure to specify the value(s) of the appropriate parameter(s)!

By a theorem from class, a linear combination of independent normally distributed RVs is also Normal.

$$E(W_1) = E(X - 2Y) = E(X) - 2E(Y) = -2 - 2(3) = -2 - 6 = \underline{\underline{-8}}.$$

$$V(W_1) = V(X - 2Y) = V(X) + 4V(Y) = 9 + 4(4) = 25 \Rightarrow SD(W_1) = \sqrt{25} = \underline{\underline{5}}.$$

Hence, $W_1 \sim N(-8, 5)$.

6. (6 pts) Let $Y_1, Y_2, \dots, Y_n$ be independent continuous random variables with cdf $F(y)$ and pdf $f(y)$. Prove that

$$f_{Y_{(n)}}(y) = n[F(y)]^{n-1} \cdot f(y).$$

$$\begin{aligned} F_{Y_{(n)}}(y) &= P(Y_{(n)} \leq y) \\ &= P(Y_1 \leq y, Y_2 \leq y, \dots, Y_n \leq y) \\ &= P(Y_1 \leq y) \cdot P(Y_2 \leq y) \cdot \dots \cdot P(Y_n \leq y) \quad [\text{by independence}] \\ &= [F(y)]^n. \end{aligned}$$

Then, we can use the Chain Rule to differentiate $F_{Y_{(n)}}(y)$.

This results in $f_{Y_{(n)}}(y) = n[F(y)]^{n-1} \cdot f(y)$. $\square$